Device-Guided Calibration of the Stanford RRAM Model: Generalization, Identifiability, and Deposition Trends in Sputter-Deposited TaO_x Devices

Md Tawsif Rahman Chowdhury, Alireza Moazzeni, and Gozde Tutuncuoglu

***Abstract*—Compact models are often calibrated to RRAM DC switching data by matching one measured I–V loop. For device and process studies, however, a visual overlay is not enough: the extracted parameters must describe the measured cycle population, and the parameters used for process comparison must be repeatably determined by the data. We report a device-guided extraction flow for the Stanford–PKU RRAM model and apply it to twelve sputter-deposited TaO_x conditions spanning 20–35% oxygen and 75–250 W RF power. The flow classifies conduction regimes, uses those regimes to set physically informed priors and loss weights, fits the Verilog-A model through HSPICE-in-the-loop Bayesian optimization, and then reports two checks with every extracted parameter set: an out-of-sample cycle test and a cycle-bootstrap identifiability audit. The fitted models reproduce the measured butterfly curves with mean log-current RMSE of 0.530 decades for SET and 0.761 decades for RESET. More importantly, a model fitted to one representative cycle predicts all other measured cycles of the same device with essentially the same error (held-out medians of 0.542 and 0.800 decades for SET and RESET). This indicates that the fit represents the device behavior, not only the chosen target trace. Bootstrap resampling shows that the branch voltage scales are repeatable (CV $\approx$ 0.28) and carry deposition trends, while series resistance and velocity prefactors are not repeatable enough for physical interpretation (CV $>$ 1). The same regime analysis shows that the post-RESET high-resistance state is Schottky-dominant in 92.5–100% of cycles, with physically admissible dynamic permittivities of 4.7–32. Thus, for these TaO_x devices, reliable compact-model extraction requires three items to be reported together: fit quality, cycle-to-cycle generalization, and which extracted parameters are actually identifiable.**

***Index Terms*—RRAM, TaO_x, compact model, Stanford model, identifiability, Bayesian optimization, out-of-sample validation, conduction mechanisms, variability.**

I. INTRODUCTION

RESISTIVE random-access memory (RRAM) based on metal-oxide filamentary switching is being actively explored for embedded nonvolatile memory, neuromorphic hardware, and in-memory computing [1]–[3]. In each of these applications, device measurements eventually have to become a compact model that circuit designers can simulate and that fabrication teams can compare across process conditions. The Stanford–PKU RRAM model [4], [5] is a common choice for this purpose because it gives a compact Verilog-A description of bipolar filamentary switching through a gap state variable, a tunneling-like $\sinh$ current, and thermally accelerated gap evolution. Related physics-based and behavioral models are also available [6]–[8]. Regardless of the chosen model, the extraction problem is the same: a measured switching loop contains real device physics, cycle-to-cycle variability, and model limitations all at once.

This makes the usual calibration figure, a simulated curve overlaid on one measured I–V loop, necessary but not sufficient. A good overlay can be obtained even when different parameter sets describe the same data almost equally well. For a device engineer, this matters because the extracted numbers may then be mistaken for material or process information. For example, two fits can show nearly identical butterfly curves while assigning different values to activation energy, series resistance, or effective gap parameters. In that case the overlay is useful as a behavioral model, but the individual parameters should not automatically be interpreted as fabrication-dependent physical quantities.

In this paper we use *identifiability* in this practical sense: a parameter is useful for device interpretation only if the measured data determine it repeatably. We also separate identifiability from *generalization*. Generalization asks whether a parameter set fitted to one representative cycle predicts the other measured cycles of the same device. Identifiability asks which fitted coordinates remain stable when the measured cycle population is resampled. These are simple questions, but they are rarely reported with RRAM compact-model parameter sets.

We address this gap using sputter-deposited TaO_x RRAM devices fabricated across a deposition design of experiments. Oxygen fraction and RF sputter power are varied over twelve sample conditions, giving a natural test case: after fitting the Stanford model, which extracted quantities can actually be tracked versus deposition? Our answer is deliberately conservative. We first identify the dominant conduction regimes in the measured curves, then use those regimes to guide the fitting, and finally report whether the fitted model generalizes across cycles and which parameters are repeatable enough to support process trends. A Fisher/Cramér–Rao calculation is retained only as a local sensitivity check; the main identifiability evidence comes from cycle-level bootstrap resampling of the measured data.

Manuscript received June 17, 2026. (*Corresponding author: M. T. R. Chowdhury.*)

The authors are with the Department of Electrical and Computer Engineering, Wayne State University, Detroit, MI 48202 USA.

The contributions are:

1) *A device-first fitting flow.* Each resistance-state segment is classified as ohmic, Poole–Frenkel (PF), Schottky, or Fowler–Nordheim (FN) before compact-model fitting. The classification is used both to set physically informed prior windows and to flag portions of the data that the released Stanford current law cannot directly represent.
2) *A direct cycle-to-cycle generalization test.* The model is fitted to one representative measured cycle, then scored against every other measured cycle from the same device. The held-out RMSE matches the fitted-cycle RMSE to within 0.04 decades on average, showing that the extracted parameter set describes the device population rather than memorizing one trace.
3) *A practical identifiability audit.* We use 200 cycle-bootstrap refits per condition to determine which parameters are repeatable under measured cycle variability. Fisher sensitivity is reported as a supporting local diagnostic, not as the final basis for physical interpretation.
4) *Deposition-resolved interpretation.* Across the twelve deposition conditions, only the repeatable branch voltage scales carry reliable process trends. Series resistance, velocity prefactors, and several gap-related coordinates can fit the data but are not stable enough to be treated as deposition-dependent physical outputs. The high-resistance state after RESET is also Schottky-limited in nearly all cycles, identifying a device-physics mechanism outside the Stanford bulk current law.

Section II describes the devices and measurement design; Section III summarizes the extraction pipeline; Section IV reports regimes, fits, generalization, identifiability, ablations, and deposition-resolved trends; Section V discusses implications; and Section VI concludes.

II. Devices and Measurement Design

The measured devices are sputter-deposited TaO_x cross-point RRAM cells with a $4\,\mu m$ feature size. They were fabricated and measured as part of a deposition-condition study [19], [20]; the asymmetric Ta-oxide bilayer stack is consistent with the established TaO_x RRAM device family [15], [21]. The experimental variables are intentionally fabrication-facing: oxygen percentage in the sputter ambient and RF sputter power. Twelve sample conditions (S1–S12) form a face-centered factorial design with four replicated center points (Table I). This layout lets us ask whether any extracted compact-model parameter is stable enough to be plotted as a process trend.

Bipolar DC switching was measured using a Keysight B1500 semiconductor parameter analyzer. The raw CSV files contain SET, RESET, or mixed sweep blocks. The fitting current compliance is $I_{cc} = 5 \times 10^{-4}$ A; because the public Stanford Verilog-A model does not include the external instrument compliance circuit, simulated currents are clamped to this value after each transient. Before fitting, an automated quality screen removes shorted, open, incomplete, or noise-floor sweeps using point count, voltage span, peak current, dynamic range, and SET+RESET completeness. For the representative S1 device, all 40 measured cycles pass the screen with a mean quality score of 8.9/10.

TABLE I
Deposition design of experiments. Center point (27.5%, 162.5 W) is replicated four times (S9–S12).

Sample	O_2 (%)	Power (W)
S1	20	75
S2	20	250
S3	35	75
S4	35	250
S5	20	162.5
S6	35	162.5
S7	27.5	75
S8	27.5	250
S9–S12	27.5	162.5

III. Extraction Methodology

The extraction flow is summarized below as eight steps. The intent is to keep the model calibration tied to the measured device behavior: select a representative cycle, identify the dominant conduction regimes, fit the Stanford model, and then test whether the fitted parameter set is useful beyond that one cycle. The released implementation contains the full prior table and all numerical constants.

A. Representative Medoid Cycle

For each condition, the fitted device is chosen by ranked quality criteria: number of good cycles, mean quality score, and good-cycle fraction. This avoids selecting a visually attractive target by hand. The SET and RESET segments are then isolated by splitting the voltage trace at direction reversals. Each cycle is interpolated in $\log_{10}|I|$ onto a common voltage grid, and the cycle closest to the branch-wise median response is selected by

$$s_{c,b} = \mathrm{median}_v \left|\log_{10}|I_{c,b}(v)| - \mathrm{median}_{c'} \log_{10}|I_{c',b}(v)|\right|, \tag{1}$$

averaged over both branches. We call this measured cycle the *medoid*. Using an actual measured sweep, instead of a synthetic median curve, preserves the hysteresis order, switching transition, and compliance-limited segments that the compact model must reproduce. The other measured cycles are not discarded; they are reserved for the out-of-sample test (Section IV-C) and the bootstrap uncertainty analysis (Section III-H).

B. Conduction-Regime Classification

Before fitting the compact model, we ask what type of conduction is visible in each measured resistance-state segment. The medoid SET and RESET branches are split at their voltage extrema into pre- and post-switching segments, corresponding to the HRS and LRS sides of each sweep. Within sliding voltage windows, four standard transport linearizations are tested:

$$\text{ohmic:} \quad \ln I = m \ln V + b, \quad 0.8 \le m \le 1.2, \tag{2}$$

$$\text{PF:} \quad \ln(I/V) = s_{\mathrm{PF}}\sqrt{V} + b, \quad s_{\mathrm{PF}} > 0, \tag{3}$$

$$\text{Schottky:} \quad \ln I = s_{\mathrm{S}}\sqrt{V} + b, \quad s_{\mathrm{S}} > 0, \tag{4}$$

$$\text{FN:} \quad \ln(I/V^2) = s_{\mathrm{FN}}/V + b, \quad s_{\mathrm{FN}} < 0. \tag{5}$$

PF and Schottky plots can both look nearly linear over a limited voltage range [13]. To avoid assigning a mechanism only because a line fits well, the fitted $\sqrt{V}$ slope s is first converted into an effective dynamic relative permittivity,

$$\varepsilon_r^{\mathrm{dyn}} = \frac{q^3}{\pi\varepsilon_0 t_{\mathrm{ox}}(skT)^2} \times \begin{cases} 1, & \text{PF}, \\ 1/4, & \text{Schottky}, \end{cases} \tag{6}$$

and must fall in the deliberately broad bracket $[1, 60]$ for $t_{\mathrm{ox}} = 7\,\mathrm{nm}$ and $T = 300\,\mathrm{K}$. Candidates that pass this physical check are compared using the Bayesian information criterion, and adjacent windows with the same label are merged and refit. Ohmic and PF windows are marked `stanford_valid`, because they are the closest to what the released Stanford current law can emulate. Schottky, FN, and unclassified windows are still reported, but they are treated as regions where the compact model may have a structural mismatch. The same classification is applied to every measured cycle so that mechanism assignments can be checked for cycle-to-cycle stability.

C. Compact Model and HSPICE Forward Map

The compact model fitted in this work is the Stanford–PKU RRAM Verilog-A model `rram_v_1_0_0` [4], [5]. Each candidate parameter set is evaluated by generating an HSPICE transient netlist and replaying the measured voltage waveform. The model current and gap evolution are

$$I_{\mathrm{tb}} = I_0 \exp\left(-\frac{g}{g_0}\right) \sinh\left(\frac{V_{\mathrm{tb}}}{V_0}\right), \tag{7}$$

$$\frac{dg}{dt} = -\nu_0 \exp\left(-\frac{qE_a}{kT_{\mathrm{cur}}}\right) \sinh\left(\frac{\gamma a_0}{t_{\mathrm{ox}}}\frac{qV_{\mathrm{tb}}}{kT_{\mathrm{cur}}}\right), \tag{8}$$

where $T_{\mathrm{cur}} = T_{\mathrm{ini}} + |V_{\mathrm{tb}} I_{\mathrm{tb}} R_{\mathrm{th}}|$ and γ is the gap-dependent field-enhancement factor. A series resistor R_s represents probe and parasitic resistance. Every objective evaluation runs two simulations: a SET sweep initialized in the HRS and a RESET sweep initialized in the LRS. Parameters that naturally differ by polarity (V_0, E_a, $F_{\min}$, ν_0, $g_{\max}$, and g_{ini}) are allowed to have SET and RESET values, while scale and parasitic parameters remain shared. Because the voltage samples are replayed quasi-statically, fitted kinetic quantities such as ν_0 and E_a include the effective sweep-time scale of this measurement protocol and should not be treated as protocol-independent material constants.

D. Regime-Conditioned Priors

Each of the 21 model parameters is assigned a prior mean, prior standard deviation, and hard physical bounds. These priors are not meant to force a particular answer; they keep the optimizer in a physically reasonable region when the DC data do not constrain every coordinate. When a regime map is available, the prior windows are tied to the classified parts of the measured curve. The LRS current scale (I_0, V_0) is estimated from $I = A\sinh(V/V_0)$ over the ohmic LRS window. If a PF window is actually present, its slope and intercept are used to inform γ_0 and E_a. If no PF window exists, the code does not force PF physics onto non-PF data; it keeps literature-centered defaults ($\gamma_0 = 16.5$, $E_a = 0.6\,\mathrm{eV}$) with larger uncertainty and records the missing PF evidence through a `pf_available` flag. This distinction is important for the TaO_x devices studied here, where most HRS segments are Schottky-like rather than PF-like. The series-resistance prior is sized from the measured LRS resistance.

E. Mechanism-Aware Objective

The optimizer compares simulated and measured currents on a logarithmic scale, because the sweep spans multiple current decades. The total objective combines branch errors, switching-threshold error, and a soft prior penalty:

$$L(\boldsymbol{\theta}) = L_{\mathrm{SET}} + L_{\mathrm{RESET}} + 0.5L_V + 0.1L_{\mathrm{prior}}, \tag{9}$$

where each branch loss is a weighted mean of squared residuals $r_i = \left(\log_{10}|I_i^{\mathrm{sim}}| - \log_{10}|I_i^{\mathrm{meas}}|\right)/\sigma_i^{\log I}$. The term L_V penalizes mismatch in the switching thresholds, identified from extrema of $d\log_{10}|I|/dV$, and L_{prior} penalizes excursions far from the prior in standardized coordinates. Points that are only covered by Schottky or FN windows receive weight $w_{\mathrm{nb}} = 0.3$, while ohmic, PF, and unclassified points receive full weight. This prevents interface-limited HRS conduction, which is not in the Stanford current law, from dominating the fitted bulk-model parameters. The unweighted RMSE is still reported later so that the residual mismatch remains visible.

F. Global Search and Local Refinement

The search itself uses Bayesian optimization over the prior-scaled parameter boxes, with log transforms for scale-like quantities [16], [17]. The configured run uses 48 quasi-random starting evaluations, 160 adaptive evaluations with an expected-improvement-plus acquisition function, and a fixed random seed; 16 of the 21 parameters are active by default. The best point is then polished by Nelder–Mead in log-parameter coordinates for 100 evaluations. A nominal point estimate therefore requires $2(48 + 160 + 100) = 616$ HSPICE branch transients, counting separate SET and RESET simulations.

G. Operational Identifiability and Generalization

After fitting, two questions are asked before assigning meaning to a parameter set. First, does the fitted model generalize to cycles it never saw during calibration? We answer this by scoring the simulated medoid fit against every other measured cycle from the same device (Section IV-C). This produces a held-out RMSE distribution for each condition. Second, which fitted parameters are stable under the measured cycle variability? We answer this primarily with the cycle bootstrap described in Section III-H: if a parameter changes substantially when the measured cycles are resampled, we do not treat it as operationally identifiable.

We also compute a finite-difference Fisher matrix $F = \widetilde{S}^{\mathsf{T}}\widetilde{S}$ in log-parameter coordinates, with effective uncertainties $\sigma_j^{\text{eff}} = \sqrt{[F^{+}]_{jj}}$ from the pseudo-inverse [9], [11]. In this manuscript the Fisher calculation is used as a local sensitivity check, not as the final identifiability decision. The reason is practical: the Fisher matrices are extremely ill-conditioned for this model and data set, so they are reliable for showing that the parameterization is broadly underconstrained but not for certifying individual physical parameters by themselves.

H. Cycle-to-Cycle Variability

Cycle-to-cycle uncertainty is estimated using 200 cluster-bootstrap draws over whole measured cycles of the selected device. In each draw, cycles are sampled with replacement, collapsed into one median SET/RESET curve by branch and point index, and refitted with a reduced optimization budget ($8+24$ evaluations) [18]. Resampling whole cycles preserves the switching sequence instead of treating neighboring voltage points as independent data. The resulting 95% intervals describe cycle-to-cycle variability for the selected device under this fitting protocol. They should not be read as wafer-level or lot-level confidence intervals.

I. Ablation Design

Finally, we fit each condition under controlled variants to see what the individual ingredients contribute: physics priors with Bayesian optimization (`physics_bo`), uninformative priors over the same hard bounds (`plain_bo`), physics-prior means without optimization (`physics_nobo`), and the full regime-conditioned flow (`physics_regime_bo`). The ablation is evaluated using *unweighted* validation RMSE, so the mechanism-aware training weights do not give the proposed method an automatic scoring advantage.

IV. Results

A. Conduction Regimes: Ohmic LRS, Schottky-Limited HRS

The first result is a device-physics result, not an optimization result: the LRS behaves as an ohmic filament branch, while the post-RESET HRS is interface-limited for essentially the entire data set. Figure 1 shows the classified regime map for the representative S1 device, and Fig. 2 summarizes the dominant cycle-level assignment for all twelve conditions. The post-SET LRS is ohmic-dominant for every condition, with assignment stability from 66.3% (S2) to 100%. The post-RESET HRS is Schottky-dominant for every condition in 92.5–100% of cycles. The extracted Schottky slopes imply $\varepsilon_r^{\text{dyn}} = 4.72$–$33.96$ for $t_{\text{ox}} = 7\,\text{nm}$, which lies inside the broad physical admissibility interval used by the classifier.

The pre-SET HRS is also Schottky-dominant in ten conditions, with stability from 60.4% (S2) to 100% (S8). S5 and S7 are exceptions and are ohmic-dominant at 80% and 75%, respectively. The pre-RESET LRS is the least stable segment and is often unclassified, consistent with the device

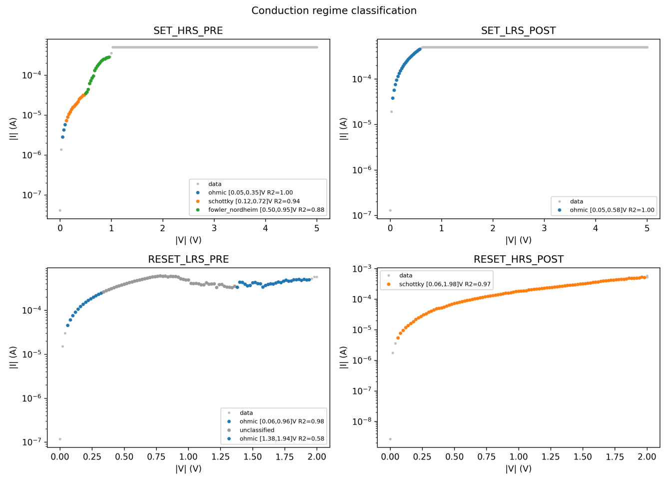

Fig. 1. Conduction-regime classification of the four resistance-state segments of the S1 representative cycle. LRS segments are ohmic ($R^2 \geq 0.98$); HRS segments are dominated by Schottky emission, with a Fowler–Nordheim window near the SET transition. No Poole–Frenkel window is admitted by the $\varepsilon_r^{\text{dyn}}$ test.

moving from a conductive-filament branch toward rupture. PF windows appear only in the representative pre-SET maps of S6 and S11; PF is not the dominant cycle-level mechanism for any state or condition.

This matters directly for model extraction. The Stanford model can reproduce the ohmic/LRS branch and switching behavior reasonably well, but its released current law does not include an explicit Schottky barrier current. Therefore, the PF-to-(γ_0, E_a) prior path is used only where PF evidence is present, and Schottky/FN HRS portions are down-weighted during fitting rather than being allowed to force bulk-model parameters into compensating for interface physics.

B. Point-Estimate Fit Quality

Table II reports the unweighted validation error for every condition. The SET branch is reproduced with RMSE from 0.47 to 0.77 decades (mean 0.530), while RESET is more difficult, with RMSE from 0.65 to 1.02 decades (mean 0.761). Visually, the model captures the hysteresis order, the main SET and RESET transitions, the compliance-limited region, and the LRS return branch. The largest residuals occur near the SET transition and along the Schottky-limited HRS approach, where the regime analysis already warned that the model current law is incomplete.

The Durbin–Watson (DW) statistics are also intentionally reported. Their values are well below 2 for all conditions, meaning the residuals are correlated along the voltage sweep rather than random point-to-point noise. For a device paper this is useful information: it says the remaining error is mostly systematic model mismatch, not measurement scatter. S2 illustrates the point, with the largest RESET RMSE (1.02 decades), the lowest SET DW, and the least stable Schottky assignment among the Schottky-dominant pre-SET conditions.

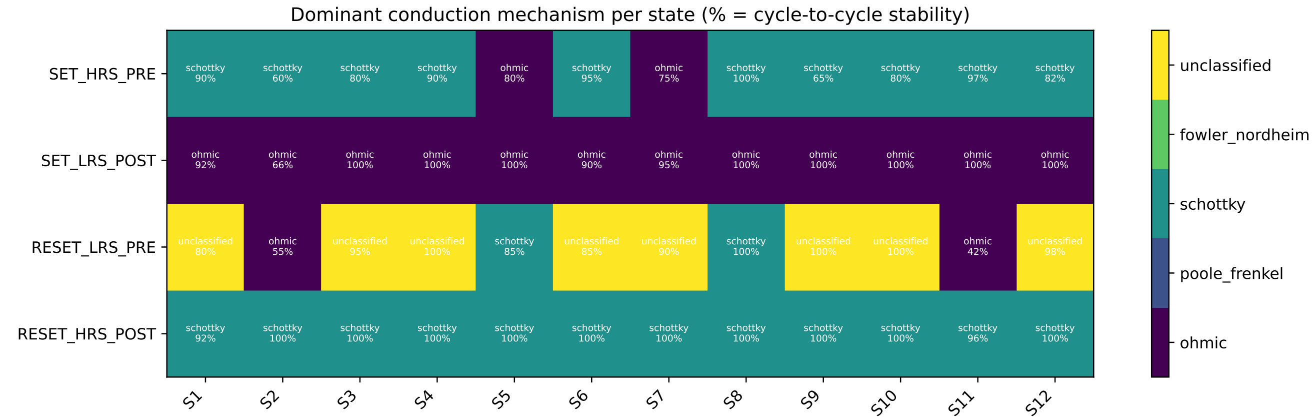

Fig. 2. Dominant conduction mechanism and cycle-to-cycle assignment stability for each resistance-state segment and deposition condition. The post-SET LRS is consistently ohmic, whereas the post-RESET HRS is Schottky-dominant for all twelve conditions.

TABLE II
VALIDATION FIT QUALITY PER CONDITION: UNWEIGHTED RMSE OF $\log_{10}|I|$ (DECADES) AND DURBIN–WATSON STATISTICS.

Cond.	$\mathrm{RMSE}_{\mathrm{SET}}$	$\mathrm{RMSE}_{\mathrm{RESET}}$	$\mathrm{DW}_{\mathrm{SET}}$	$\mathrm{DW}_{\mathrm{RESET}}$
S1	0.500	0.689	0.86	0.85
S2	0.772	1.016	0.46	0.56
S3	0.566	0.659	0.72	0.98
S4	0.514	0.730	0.82	0.90
S5	0.508	0.805	0.85	0.74
S6	0.468	0.650	0.94	0.93
S7	0.477	0.726	1.04	0.91
S8	0.495	1.001	1.03	0.54
S9	0.510	0.700	0.92	1.04
S10	0.508	0.718	1.00	0.98
S11	0.494	0.753	0.92	0.88
S12	0.548	0.681	0.96	1.01
Mean	0.530	0.761	–	–

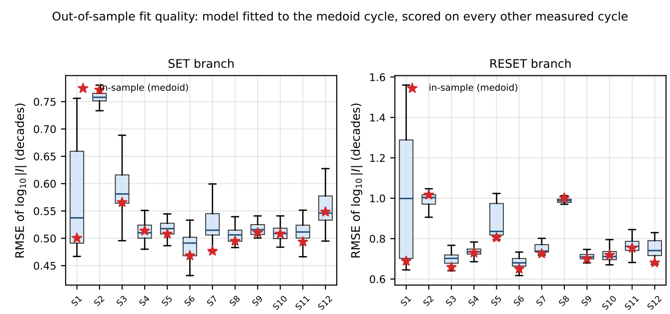

Fig. 3. Out-of-sample fit quality. Box plots: distribution of $\log_{10}|I|$ RMSE when the model fitted to the medoid cycle is scored against every other measured cycle of the device (whiskers exclude outliers). Stars mark the in-sample (medoid) RMSE. Held-out and in-sample errors are statistically indistinguishable, evidencing generalization rather than overfitting.

C. Out-of-Sample Generalization Across Measured Cycles

A parameter set fitted to one cycle is useful for circuit simulation only if it describes the other cycles from the same device. We test this directly. The simulated medoid fit is scored point-by-point on the common voltage grid against every measured cycle, not only against the medoid used during calibration. The medoid cycle gives the in-sample RMSE in Table II; the remaining 19–100 cycles per condition form a held-out cycle population.

Figure 3 shows the resulting per-cycle RMSE distributions. Averaged over all conditions, the held-out median RMSE is 0.542 decades for SET and 0.800 decades for RESET, compared with in-sample means of 0.530 and 0.761 decades. The average generalization gap is therefore only 0.012 decades for SET and 0.039 decades for RESET. For every condition, the in-sample value falls inside the held-out interquartile range. This is the key evidence that the fitted compact model represents the measured device population rather than memorizing one selected trace.

Figure 4 shows the same idea for S1 in the time/order of the sweep. The fitted curve follows the measured median and stays inside the cycle-to-cycle envelope for most of the trajectory. The departures occur at the SET transition and on the HRS approach. Across conditions, the fitted SET curve lies inside the measured min–max band for 77% of grid points on average, compared with 63% for RESET. The lower RESET coverage is consistent with the missing Schottky-limited HRS branch identified earlier, not with overfitting, because RESET is elevated both in-sample and held out.

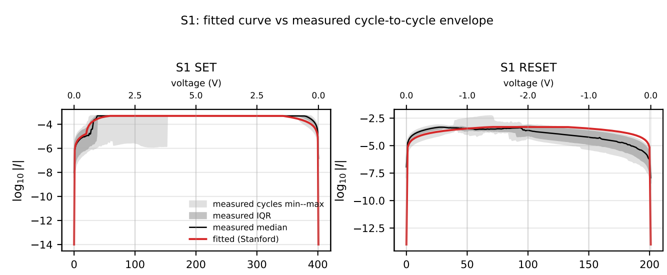

Fig. 4. S1 fitted curve (red) against the measured cycle-to-cycle envelope (min–max and interquartile bands, with the measured median) along the sweep sequence; the secondary axis marks voltage. The fit stays within the measured spread except at the SET transition and the HRS approach.

D. Identifiability: What DC Data Actually Constrain

The bootstrap answers a practical extraction question: if the measured cycle population is sampled again, do we recover the same parameter? In Fig. 5, each entry is the coefficient of variation (CV) of the bootstrap distribution for one parameter and one condition. All 2400 refits (200 draws $\times$ 12 conditions) completed, and all 192 condition–parameter intervals are finite and nondegenerate. The median CV over all entries is 0.448, with a range of 0.202–2.698.

The clearest repeatable quantities are the branch voltage scales, with median CV near 0.28. This is physically reasonable: the voltage scale controls the curvature of the measured branch, so the DC butterfly curve directly constrains it. By contrast, R_s has a median CV of 1.444, and the SET and RESET velocity prefactors have median CVs of 1.194 and 1.282. The current scale I_0 and activation energies also vary substantially. These parameters can help the optimizer fit a trace, but the DC data do not recover them repeatably enough for process interpretation.

The Fisher analysis supports the same broad conclusion but should be read with caution. Across conditions, the Fisher condition numbers range from 6.2×10^{18} to 1.6×10^{66}, far beyond what can support precise per-parameter certification in double precision. Figure 6 therefore serves as a local sensitivity map rather than the main identifiability decision. Where Fisher and bootstrap agree, the conclusion is strong: branch voltage scales and switching-field combinations are better determined than I_0, g_0, activation energies, velocity prefactors, and gap-geometry parameters. Where they disagree, the bootstrap is more important for this device study. The best example is R_s, which can look locally well determined at one optimum yet varies strongly under cycle resampling. A subset of profile-likelihood scans (S1–S4) also returns "flat/non-identifiable" behavior for nearly all coordinates, consistent with the bootstrap reading.

The device-level interpretation is therefore narrow but useful: DC butterfly data constrain effective branch scales and switching-field combinations more reliably than they constrain a unique decomposition into current prefactors, activation energies, gap geometry, and kinetic prefactors.

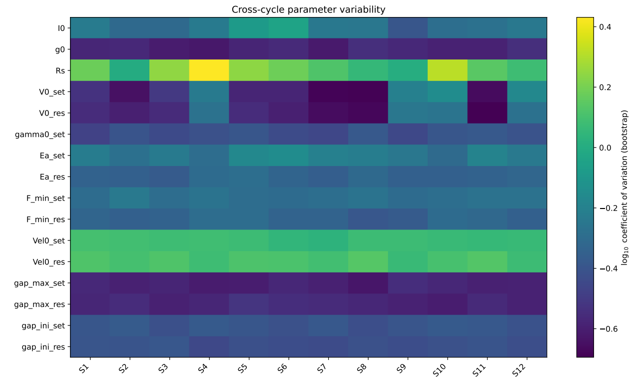

Fig. 5. Cycle-to-cycle parameter variability from 200 cluster-bootstrap refits per condition (color: $\log_{10}$ CV). Branch voltage scales are comparatively repeatable; series resistance and velocity prefactors are the least stable. This bootstrap CV is our primary identifiability evidence.

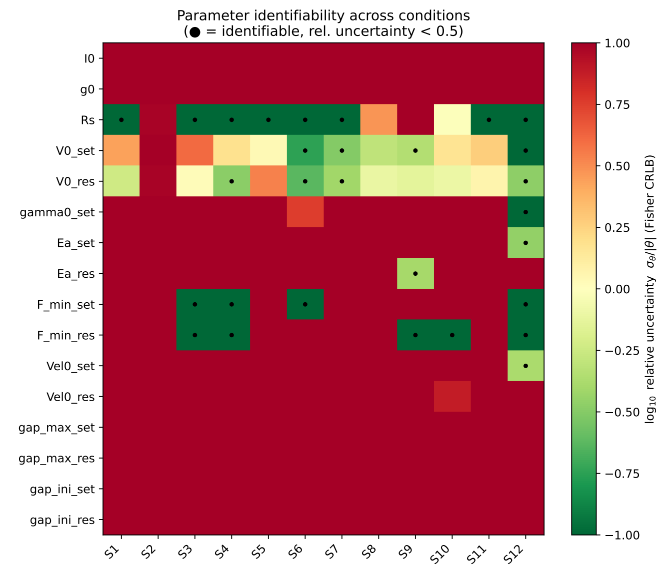

Fig. 6. Local Fisher relative uncertainty $\sigma_\theta/|\theta|$ (log color scale) for the 16 active parameters across the twelve conditions, shown as a corroborating cross-check. The extremely large condition numbers (10^{18}–10^{66}) preclude certifying individual coordinates from this diagnostic alone; conclusions are drawn only where it agrees with Fig. 5.

E. Ablation: What Each Ingredient Buys

Table III compares the fitting variants using unweighted validation RMSE. The most obvious requirement is optimization itself. On RESET, using physics-informed prior means without Bayesian optimization gives 1.528 ± 0.434 decades, while optimized variants are near 0.756 ± 0.124 decades. SET is less sensitive because the LRS and compliance-limited portions are already close to the prior prediction.

The three optimized variants have similar aggregate RMSE. This is informative because regime conditioning and informative priors do not mainly buy a lower overlay error; they keep the fit from using unsupported Schottky/FN HRS regions to distort parameters that belong to the Stanford bulk current law. The ablation therefore reinforces the main message: a compact-model overlay alone is not enough to establish physical parameter extraction. Generalization and identifiability are the quantities that decide what can be trusted.

TABLE III
ABLATION OVER ALL 12 CONDITIONS: UNWEIGHTED VALIDATION RMSE (DECADES), MEAN $\pm$ STANDARD DEVIATION ACROSS CONDITIONS.

Variant	RMSE_{SET}	$\text{RMSE}_{\text{RESET}}$
Regime-conditioned + weighting	0.530 ± 0.081	0.761 ± 0.123
Physics priors + BO	0.546 ± 0.077	0.756 ± 0.124
Uninformative priors + BO	0.537 ± 0.076	0.755 ± 0.126
Physics priors, no BO	0.571 ± 0.115	1.528 ± 0.434

F. Deposition-Resolved Extraction

The deposition DOE in Table I gives a direct way to use the identifiability result. A response surface should only be built

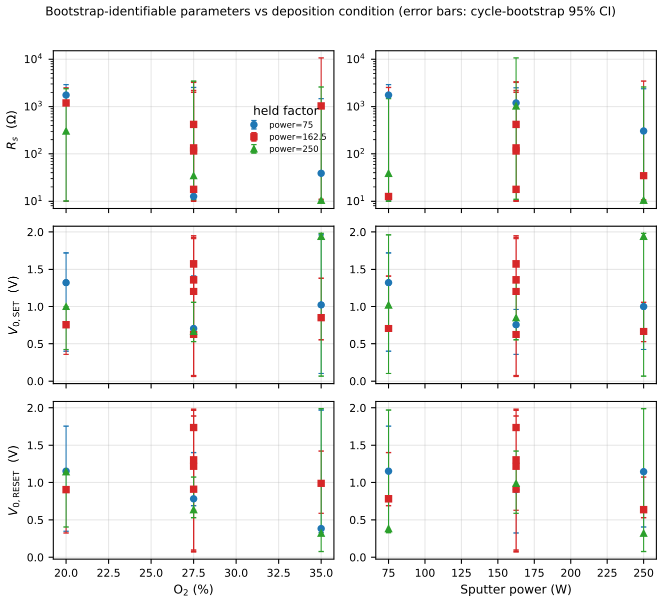

Fig. 7. Bootstrap-identifiable parameters versus deposition condition (error bars: cycle-bootstrap 95% CI). Branch voltage scales (middle, bottom rows) are tightly constrained and carry deposition information; series resistance (top) spans up to two decades and cannot support a response surface despite being locally Fisher-identifiable.

from parameters that are repeatably extracted; otherwise the apparent process trend may simply be fit ambiguity. Figure 7 compares these two cases. The branch voltage scales $V_{0,\mathrm{SET}}$ and $V_{0,\mathrm{RESET}}$ have relatively tight bootstrap confidence intervals and vary systematically with deposition. For example, $V_{0,\mathrm{RESET}}$ tends to be lower at the 35% oxygen, low-power corners (S3 and S4) than at the oxygen-poor corners (S1 and S2). These quantities can therefore support deposition-resolved compact-model interpretation.

Series resistance gives the opposite lesson. It is often locally sensitive in the Fisher calculation, but its bootstrap 95% intervals span one to two orders of magnitude. Its apparent changes across conditions cannot be separated from cycle-to-cycle extraction uncertainty. For a fabrication or process study, R_s should therefore not be used as a response-surface output from these DC sweeps, even if a single fitted curve appears to prefer a particular value.

The regime classifier also provides a process-facing HRS observable that is not a Stanford-model fitting parameter. Figure 8 plots the dynamic relative permittivity inferred from the post-RESET Schottky window. All twelve conditions fall in the physically admissible range of 4.7–32, supporting the conclusion that the post-RESET HRS is interface-limited across the entire DOE. The approximately three-fold scatter among the replicated center-point devices (S9–S12) further suggests that the interface response has its own device-to-device and cycle-to-cycle variability, not just a single-valued dependence on the nominal deposition setpoint. This is an automated, quantitative extension of mechanism analysis that was previously performed manually [19].

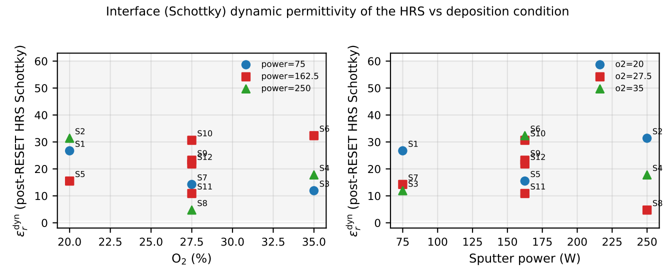

Fig. 8. Dynamic relative permittivity inferred from the post-RESET HRS Schottky window for all twelve conditions, versus oxygen (left) and power (right). Shaded band: admissibility interval $[1, 60]$. All conditions are physically admissible, corroborating interface-limited HRS conduction device-wide.

V. Discussion

A. Implications for Compact-Model Users

For circuit users, the result is encouraging but bounded. The calibrated Stanford parameter files are useful behavioral descriptions for the measured DC protocol: a fit obtained from one representative cycle predicts the other cycles of the same device with essentially the same error. This supports their use in circuit-level studies that need a compact description of the observed switching loops.

For device and fabrication interpretation, the message is more selective. Only repeatable parameter combinations should be compared across deposition conditions. In this data set, the branch voltage scales and related switching-field combinations pass that test. Fitted E_a, ν_0, gap geometry, and even R_s do not have enough cycle-bootstrap repeatability to be treated as standalone physical outputs from DC sweeps. Reporting the full parameter table is still useful for reproducibility, but the process discussion should be based on the identifiable subset.

B. Model Adequacy for TaO_x

The regime analysis also explains why RESET remains harder to fit than SET. The post-RESET HRS is Schottky-dominant in nearly all cycles, while the released Stanford model uses a single bulk-like $\sinh$ current law. The remaining RESET error is therefore a model-adequacy issue, not simply an optimizer issue. This interpretation is supported by the ablation study, where the optimized variants reach similar unweighted RMSE, and by the held-out test, where RESET error is elevated both in-sample and out of sample.

A natural next model extension would add a barrier-limited HRS branch or a gap-dependent barrier term [6], [7], [13]. The regime map gives the experimental windows and slopes that such an extension should target. Until then, the present flow keeps the Schottky mismatch visible in the unweighted validation metrics while preventing those regions from dominating extraction of parameters associated with the Stanford bulk law.

C. Limitations

Several limitations should be kept explicit. First, the transient replay is quasi-static, so kinetic parameters absorb the effective measurement time scale; pulsed or sweep-rate-resolved data would be needed to constrain kinetics more directly [14]. Second, the bootstrap and out-of-sample analyses are cycle-level analyses for one selected device per condition. They quantify cycle-to-cycle variability under the fitting protocol, not full device-to-device or wafer-level variability, although the replicated center points give an initial view of that spread. Third, Fisher analysis is used only as a local cross-check, and profile-likelihood scans were completed for a subset of conditions (S1–S4); a future release should complete those scans for all conditions using a common optimum. Finally, the reset-side γ_0 is structurally overridden under negative bias in the released Verilog-A model, so it is retained for compatibility but not physically interpreted.

VI. Conclusion

We presented a device-guided extraction flow for the Stanford RRAM compact model and applied it to twelve sputter-deposited TaO_x deposition conditions. The fitted models reproduce the measured DC switching loops with mean RMSE of 0.530 decades for SET and 0.761 decades for RESET. More importantly, the fits generalize across measured cycles: when a model fitted to one representative cycle is scored against all other cycles from the same device, the held-out median RMSE remains close to the in-sample value (0.542/0.800 decades for SET/RESET).

The extracted parameter table should not be interpreted uniformly. Cycle bootstrap shows that branch voltage scales are repeatable and carry usable deposition trends, while series resistance, velocity prefactors, activation energies, and several gap-related parameters are not repeatable enough for standalone physical interpretation from these DC sweeps. The regime analysis adds the corresponding device-physics reason for the residual floor: the post-RESET HRS is Schottky-dominant in 92.5–100% of cycles, with admissible dynamic permittivities, while the released Stanford current law does not explicitly include that interface-limited branch. For RRAM compact-model calibration, fit quality should therefore be reported together with cycle-level generalization, parameter identifiability, and conduction-regime validity.

References

[1] H.-S. P. Wong *et al.*, “Metal-oxide RRAM,” *Proc. IEEE*, vol. 100, no. 6, pp. 1951–1970, Jun. 2012.

[2] D. Ielmini and H.-S. P. Wong, “In-memory computing with resistive switching devices,” *Nature Electron.*, vol. 1, pp. 333–343, 2018.

[3] S. Yu, “Neuro-inspired computing with emerging nonvolatile memories,” *Proc. IEEE*, vol. 106, no. 2, pp. 260–285, Feb. 2018.

[4] X. Guan, S. Yu, and H.-S. P. Wong, “A SPICE compact model of metal oxide resistive switching memory with variations,” *IEEE Electron Device Lett.*, vol. 33, no. 10, pp. 1405–1407, Oct. 2012.

[5] Z. Jiang *et al.*, “A compact model for metal–oxide resistive random access memory with experiment verification,” *IEEE Trans. Electron Devices*, vol. 63, no. 5, pp. 1884–1892, May 2016.

[6] M. Bocquet *et al.*, “Robust compact model for bipolar oxide-based resistive switching memories,” *IEEE Trans. Electron Devices*, vol. 61, no. 3, pp. 674–681, Mar. 2014.

[7] C. Bengel *et al.*, “Variability-aware modeling of filamentary oxide-based bipolar resistive switching cells using SPICE level compact models,” *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 67, no. 12, pp. 4618–4630, Dec. 2020.

[8] S. Kvatinsky *et al.*, “VTEAM: A general model for voltage-controlled memristors,” *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 62, no. 8, pp. 786–790, Aug. 2015.

[9] M. K. Transtrum *et al.*, “Perspective: Sloppiness and emergent theories in physics, biology, and beyond,” *J. Chem. Phys.*, vol. 143, no. 1, Art. no. 010901, 2015.

[10] R. N. Gutenkunst *et al.*, “Universally sloppy parameter sensitivities in systems biology models,” *PLoS Comput. Biol.*, vol. 3, no. 10, pp. 1871–1878, 2007.

[11] A. Raue *et al.*, “Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood,” *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009.

[12] O.-T. Chis, J. R. Banga, and E. Balsa-Canto, “Structural identifiability of systems biology models: A critical comparison of methods,” *PLoS ONE*, vol. 6, no. 11, Art. no. e27755, 2011.

[13] F.-C. Chiu, “A review on conduction mechanisms in dielectric films,” *Adv. Mater. Sci. Eng.*, vol. 2014, Art. no. 578168, 2014.

[14] D. Ielmini, “Resistive switching memories based on metal oxides: Mechanisms, reliability and scaling,” *Semicond. Sci. Technol.*, vol. 31, no. 6, Art. no. 063002, 2016.

[15] D. J. Wouters, R. Waser, and M. Wuttig, “Phase-change and redox-based resistive switching memories,” *Proc. IEEE*, vol. 103, no. 8, pp. 1274–1288, Aug. 2015.

[16] J. Snoek, H. Larochelle, and R. P. Adams, “Practical Bayesian optimization of machine learning algorithms,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2012, pp. 2951–2959.

[17] B. Shahriari *et al.*, “Taking the human out of the loop: A review of Bayesian optimization,” *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, Jan. 2016.

[18] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall/CRC, 1994.

[19] M. T. R. Chowdhury, A. Moazzeni, and G. Tutuncuoglu, “Atomic-scale insights into the switching mechanisms of RRAM devices,” in *Proc. IEEE 68th Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Lansing, MI, USA, 2025, pp. 518–522.

[20] A. Moazzeni, M. T. R. Chowdhury, and G. Tutuncuoglu, “The impact of operation conditions on potentiation, depression and endurance dynamics of tantalum oxide RRAMs,” in *Proc. IEEE Int. Integr. Rel. Workshop (IIRW)*, 2023.

[21] M.-J. Lee *et al.*, “A fast, high-endurance and scalable non-volatile memory device made from asymmetric Ta_2O_{5-x}/TaO_{2-x} bilayer structures,” *Nature Mater.*, vol. 10, pp. 625–630, 2011.